Vibhas Tallapalli

vtallapa@uwaterloo.ca | vibhastallapalli.github.io | linkedin.com/in/vibhas-tallapalli | github.com/vibhastallapalli | +1 (647) 975-6479

Education

University of Waterloo — Waterloo, ON
BASc in Nanotechnology Engineering — *Sep 2025 – Present*

Technical Skills

Microcontrollers: Arduino, ARM, AVR, Embedded Systems, Serial Communication, Sensor Integration
Mechanical: SolidWorks, Onshape, CAD, GD&T, Tolerance, DFM/DFA, Carbon Fiber Fabrication, Mill, Lathe, Bandsaw
Software: C++, Python, C, MATLAB, Git, Arduino C++, PID Control, Matplotlib, Tkinter, PPG, IMU

Experience

Linear Automation Inc. | *SolidWorks, FEA, GD&T, MATLAB* — Barrie, ON
Mechanical Engineer Co-op — *Apr 2026 – Present*

- Designed and stress-tested **5+ structural Honda vehicle frame components** in **SolidWorks** using **MATLAB** and **FEA** across **7+ iterations**, running parameter sweeps across wall thickness and load cases to isolate stress concentration failures at oval cross-sections, then redesigning geometry and upgrading material (**Al 5000→6000 series**) to cut component weight by **30%** while maintaining **300+ lb** structural load requirements.
- Converted a **50+ hole manifold** from **metric to NPT threading**, resolving **hole sizing**, **wall clearance**, and **tapered thread engagement depth** across **6+ revisions** to ensure every port met dimensional accuracy and prevent a **$400+ scrapped part** from reaching the production floor.
- Produced **100+ production drawings** across unique parts and assemblies for **6 senior engineers**, applying **GD&T callouts**, **tolerances**, and **STEP file updates** so every part reached the shop floor ready to build, eliminating **2+ hour backtracking cycles** per bad callout sent to manufacturing.

Waterloop Hyperloop | *Onshape, SolidWorks, Machining (Mill/Lathe), Mechanical Assembly* — Waterloo, ON
Mechanical Engineer — *Sep 2025 – Jan 2026*

- Maintained a **100+ part Onshape assembly** integrating brake suspension battery packaging to coordinate mechanical design across subteams.
- Designed and machined **5 aluminum brake** and pneumatic components from SolidWorks drawings using **mill/lathe workflows**, then installed and fit-checked them on the pod to confirm alignment and clearances, saving **2 weeks of build time**.

Electrium Electric Vehicles | *SolidWorks, Mechanical Packaging, Design for Assembly, Prototyping* — Waterloo, ON
Hardware Design Engineer — *Sep 2025 – Jan 2026*

- Created a **SolidWorks CAD model** of the **F2025 e-bike** to centralize frame geometry and key dimensions (seat, handlebars, wheels), giving the team a reliable reference for fitment checks and layout decisions while speeding up enclosure/component packaging.
- Redesigned the **battery enclosure redesign** to improve service access and assembly by replacing welded aluminum with a **tempered-glass panel** and revised mounts, reducing welding/hardware requirements and cutting build cost by **$300+ cost reduction** across 3 prototypes.

IntegraYouth — Executive | *Outreach, Public Speaking, Leadership, Event Planning* — Toronto, ON
Secretary | Blog Writer — *June 2024 – Present*

- Wrote **50+ blog posts** on programs and impact, reaching **1,000+ views** and improving outreach through consistent publishing and topic planning.
- Ran Secretary-team operations and supported event logistics, communications, and scheduling for **5+ events** with **200+ attendees**.

Projects

Adaptive Focus Productivity Timer | *Arduino (C++), Python (Tkinter), Serial Communication, CAD* — Nov 2025 – Jan 2026

- Developed a **3-sensor embedded occupancy detection system** on **Arduino Uno (C++)**, implementing **10 Hz polling** and cross-checking between sensors to suppress false triggers and maintain stable real-time presence detection.
- Built a **Python Tkinter desktop application** that communicates with Arduino over **serial (9600 baud)** to manage study sessions, log **100+ focus/break events** per session, and visualize performance metrics through a real-time session history dashboard with automatic JSON data logging.
- Designed and fabricated a custom **SolidWorks enclosure** to integrate the **Arduino and sensor array** into a compact, portable unit, improving durability and enabling easy setup across different study environments.

Electric Bike Control System Simulation | *PID Control, C++, Physics Modelling, Python (Matplotlib)* — Oct 2025 – Nov 2025

- Built a **C++/Python simulation** of an e-bike speed control system, testing **50+ rider load conditions** and analyzing **100+ controller responses** to ensure the bike held a steady target speed while maximizing battery efficiency or feeling unstable when conditions change.
- Implemented and tuned a **PID speed controller** with throttle saturation and anti-windup (integral clamping), reducing **15% overshoot reduction**, **20% settling time improvement**, and improving simulated energy efficiency by 40% under changing load/terrain.

Micro Wind Energy Harvesting Turbine | *CAD (Simulation), Fabrication, Tolerance/GD&T, DC* — Sept 2025 – Oct 2025

- Designed the **SolidWorks rotor** in SolidWorks to get a tighter, balanced fit on the shaft (less wobble/friction), as well as built it using PVC and nylon through sanding/alignment adjustments, allowing the turbine to spin more smoothly and raising measured **32% DC output voltage increase**.
- Converted rotation into DC output by driving a **DC generator** from the rotor shaft and logging voltage across 10+ wind-test runs with a multimeter.